

# A Geospatial approach for E-waste Facility Locating

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**Abstract**—*This paper presents the "E-Waste Facility Locator," a web-based application designed to facilitate appropriate e-waste disposal through the utilization of modern web technologies and location-based services. People may easily and accurately locate places near them that take e-waste through the portal. It also has educational content to raise awareness about the health and environmental risks of disposing of e-waste in an improper way. To make it eco-friendly, users will be able to enter the model number of their old devices and gain credit points based on how much precious metal (like gold or silver) was "retrieved." We use APIs like Google Maps and OpenStreetMap to make sure that the location tracking is correct. The website was built using "HTML, CSS, JavaScript, Node.js, Express.js and MongoDB". We utilize a Random Forest (machine learning) model to guess the pricing of our project. This project aims to provide individuals the tools they need to make better decisions and help make the world cleaner and healthier by combining geographic information, user involvement, and environmental education.*

**Keywords**— *"HTML, CSS, JavaScript, Node.js, Express.js, MongoDB", Geolocation, Google Maps API, E-Waste Management, Sustainability.*

## I. INTRODUCTION

With electronic devices becoming increasingly important in everyday life, so has the growing problem of e-waste. Chemicals such as lead, mercury, and cadmium pose dangers to both humans and the environment. In spite of numerous government regulations and education campaigns, proper disposal of e-waste remains an ongoing problem. These challenges include lack of access to easily identifiable recycling facilities, lack of public knowledge regarding responsible e-waste disposal, and lack of incentives for doing so properly.

In order to address these issues, the E-Waste Facility Locator was developed. This locator uses a combination of location-based tracking of e-waste, user awareness, and rewards for proper disposal of e-waste. Developed using

current web technologies (HTML, CSS, jQuery/JavaScript, Node.js and MongoDB) users may enter their zip code or city and state to find the closest certified e-waste collection center to them. Users will also see educational pop-up messages about the hazardous nature of e-waste. Users can also log information about their used items and earn credit points depending upon the amount of usable content within their item(s). This article describes the development of the website including its architecture, layout, functionality, and API's used for map display and directions. The purpose of this project includes:

- A. Examination of geolocation services through use of APIs to assist consumers in finding e-waste drop off locations nearby.
- B. Presentation of geolocation services through APIs to assist consumers in finding e-waste drop off locations.
- C. Discussion of how a credit point system that rewards consumers for recycling different types of used electronics.
- D. There is also an evaluation of whether the sending of educational notifications and messages may enhance the public's perception of the importance of recycling and their habits in regards to recycling.

The ultimate objective of this study is to determine if the platform could expand, retain the user's engagement and contribute to e-waste management globally. This study will provide a link between consumer behavior and recycling facility practices and it is an effort toward attaining one of the UN's Sustainable Development Goals (SDG) number 12 for sustainable consumption and production patterns.

## II. LITERATURE REVIEWS

Most researchers have primarily researched the issues and potential solutions to the management of e-waste. Nnorom and Osibanjo [12] examined how emerging economies such as Nigeria are dealing with e-waste. They clarified that carelessly dumping e-waste can cost lives and the environment and that public education and government regulations are extremely significant. Manomivibul [10]

researched a system named Extended Producer Responsibility (EPR) in which firms are held responsible for recycling what they sell. He discovered that in countries such as Europe and America, this system compels firms to produce goods less toxic to the environment. Grant et al. [7] examined the impact of e-waste on the health of the children in developing nations. They discovered that children who reside around informal recycling facilities mostly experience respiratory ailments and delayed brain growth due to exposure to toxic substances. Widmer et al. [13] discussed the fact that recycling e-waste is extremely crucial but is not taking place quickly enough. The principal causes are poor legislation and low public awareness. Kahhat and Williams [8] contrasted the regulations of e-waste across various nations and stated that nations that have strong legislation, such as in Europe, deal much better with e-waste. Zeng et al. [5] examined why individuals in certain developing nations do not recycle their obsolete electronics. They concluded that the majority of people do not recycle since they are unaware of how it is too difficult, or they don't receive any incentives for recycling. Ogunseitan et al. [11] discussed the hazardous chemicals in e-waste, such as lead and mercury. They also discussed how to remove or decrease these toxic materials safely at the time of recycling. Cucchiella et al. [6] contrasted e-waste recycling systems in Europe and North America. They discovered that European nations recycle e-waste more successfully and recommended that North America adopt their systems. Zhou et al. [4] applied life-cycle studies to demonstrate that recycling e-waste is far more environmentally friendly than burning or burying e-waste—if executed properly. Chi et al. [9] were interested in e-waste recycling workers in countries such as India and China who worked without safety equipment or training. To enhance both safety and quality in recycling, the authors advocate for changes within formal systems.

In addition to advocating for such formal system changes, the authors' research contributed to the development of the E-Waste Facility Locator. The authors found that individuals have difficulty locating the proper facilities for disposing of their e-waste, lack awareness of where those facilities exist, and perceive no personal benefit from participating in recycling programs. Therefore, the locator platform uses location base searching, easily accessed educational resources, and incentives to provide access to recycling as well as to create informed and rewarding experiences when recycling. The platform helps build trust and provides a convenient experience for individuals to be connected with certified recycling facilities. Additionally, over time, the platform is expected to create a culture of responsibility regarding the management of e-waste among consumers. This creates a positive impact on the environment, and contributes to creating sustainable communities.

**Table I: Feature-Based Comparison of Scraply and Existing E-Waste Solutions**

| Point                                  | Scraply (Our Website)                               | Cashify                            | Karo Sambhav                           | Attero                                 |
|----------------------------------------|-----------------------------------------------------|------------------------------------|----------------------------------------|----------------------------------------|
| <b>1. Facility Locator</b>             | 3D interactive map for nearby authorized facilities | Pickup service only, no live map   | Has ops info/map, not consumer - first | Form-based locator, no interactive map |
| <b>2. Price Prediction</b>             | Random Forest model with ~95% accuracy              | AI-based quotes, model undisclosed | No public pricing engine               | No consumer price engine               |
| <b>3. Transparency</b>                 | Open model disclosure (Random Forest + parameters)  | Proprietary, black-box             | Not applicable                         | Not applicable                         |
| <b>4. Awareness</b>                    | Pop-ups + blogs educating users on e-waste harms    | Blogs only                         | Institutional awareness campaigns      | Blogs, industry briefs                 |
| <b>5. Integrated Experience</b>        | One-stop solution: Locator + Price + Education      | Trade-in focus only                | Compliance + awareness                 | Large-scale recycling only             |
| <b>6. User-Centric Design</b>          | Designed for individuals & households               | Consumer resale focus              | Enterprise / EPR first                 | Industrial recycling first             |
| <b>7. Technological Sophistication</b> | 3D geospatial + ML models                           | Proprietary AI only                | Basic maps + campaigns                 | Collection network, no AI              |

|                                  |                                                             |                                    |                                         |                                            |
|----------------------------------|-------------------------------------------------------------|------------------------------------|-----------------------------------------|--------------------------------------------|
| <b>8. Behavioral Impact</b>      | Interactive nudges for recycling behavior                   | Limited to resale motivation       | Awareness campaigns, less interactive   | Recycling compliance, not behavior-focused |
| <b>9. Research Orientation</b>   | Built for research with transparent methods                 | Business-driven, no academic focus | Compliance-driven, not research-centric | Industry-driven, not transparent           |
| <b>10. Holistic Contribution</b> | Complete framework combining tech + awareness + sustainable | Focuses on resale economics        | Focuses on EPR obligations              | Focuses on industrial recycling            |

**Source:** Author's own compilation based on analysis of various existing platforms and publicly available information.

Table I shows how Scraply compares against three other e-waste platforms in regard to Cashify, Karo Sambhav, and Attero. The analysis concludes that Scraply has superior service in 10 separate aspects. Scraply also has a unique 3D interactive facility locator, Random Forest-based price prediction with a model accuracy prediction of about 95%, and context-dependent educational elements such as pop-ups and blogs, which together provide not only a value, but also the user can do this with education on the process of e-waste awareness and mitigation; this conceptually enhances the potential for accurate value along with user education.

Scraply goes the extra mile in regard to transparency by showing its model parameters, whereas Cashify, Karo Sambhav and Attero have their values hidden in proprietary or private methodologies. Scraply is a one of a kind model when compared to Cashify (e-waste resale), Karo Sambhav (e-waste compliance), and Attero (e-waste as an industrial recyclable), because this online framework deals with technology, consumer behaviour and sustainability in a discipline-wide manner. Scraply is a legitimate research model that addresses individual and community efforts to combat problems created by e-waste technologies.

In this way, the results of the research were transformed into something useful for people and the natural world.

### III. IMPLEMENTATION

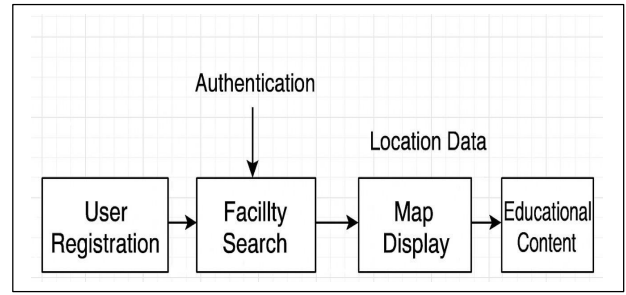


Fig. 1. System Design for E-Waste Facility Locator

The E-Waste Facility Locator will first ask users to sign up so they may use it safely. They must first authenticate their identity before searching for e-waste disposal centers. Later, the facility will present a map which shows all results from searching; the nearest and most suitable places from the location of the user are displayed. At last, it presents a part with educational information that helps users to recognize why e-waste management matters and also about how daily life can be more eco-friendly. To find out how useful this E-Waste Facility Locator is, this project outlines a systematic step-by-step method. The primary purpose is to build an easy to use system to help users locate newer e-waste recycling centers, learn the dangers of e-waste, and earn points for using proper methods of disposal and not throwing it away in the garbage. This method focuses around building a simple web app using real time locations, helpful information, and a point-based rewards system. The project will also explore existing e-waste platforms to find out what aspects are functioning well, what issues they have, and what they are doing poorly. This information will be instrumental in improving the platform to provide a better service that will be more readily usable, reliable, and accurate to everyone.

#### A. METHODOLOGY

The E-Waste Facility Locator is a fully modernized, full-stack web application that has responsive, stable, and scalable utility. This application has three main components: frontend, backend, and database to support the geolocation APIs and additional services.

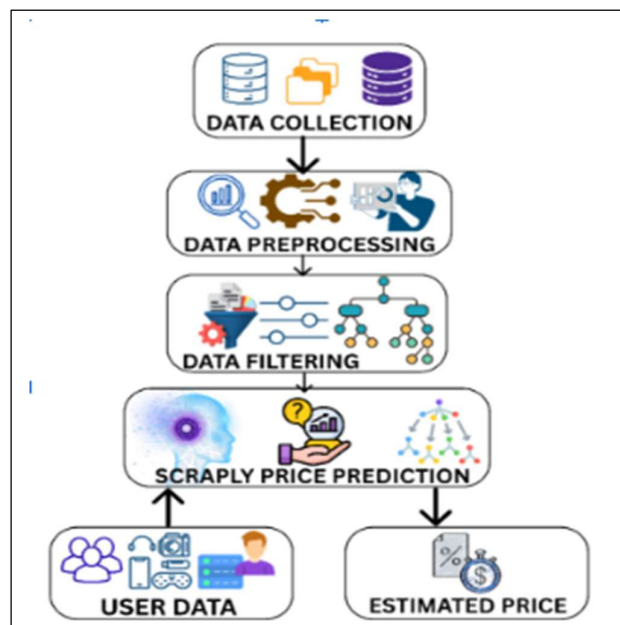
**Frontend** is designed and developed using HTML, CSS, JavaScript, and React.js. The frontend allows the user an interactive and mobile-usable application. The functionality of the front-end are user ability to search for the facilities, input a device to receive credit points, educational pop-ups, and a map display that supports visualization using **Leaflet.js** or **Mapbox**.

**Backend** is designed and developed using **Node.js** and **Express.js**. The backend has the ability to have users register, log in, receive and process data requests, and track and process users' credit points. The connection to the front-end and backend is done using RESTful APIs.

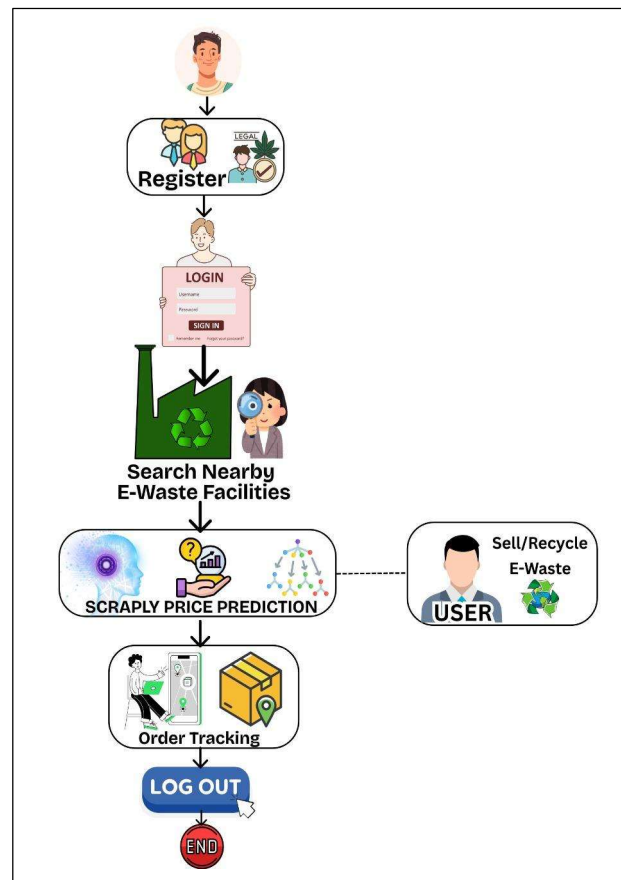
**The database** is developed with MongoDB, which is fast, flexible, and applicable to the purposes we needed. The database stores user accounts, facility user information, educational content, models of devices, and total credit points.

**APIs Used:** Google Maps API or OpenStreetMap API can be used for facility location and navigation. Firebase or Auth0 can be used for user authentication. Additionally, custom API logic can be implemented to do point calculations and user management, to ensure the system is functional and easy to use.

**Price Prediction:** In the beginning, to predict prices, we used the simplest method of a decision tree and then transitioned to the better-performing Random Forest model. The decision tree was helpful, as it provided an intuitive and clear way to show mappings in the data and was particularly useful to visualize non-linear relationships. However, it lacked performance because a decision tree is often prone to overfitting, which limits usefulness on unknown data. Because we were only using a single decision tree, being prone to overfitting meant the predictions were not reliable. To mitigate this risk, we turned to using a Random Forest which is a model that combines the rationale of many different decision trees, each trained on a random sample of the data and with only a random set of features considered at every split, meaning each of the trees ended up being less correlated with each other. By averaging the predictions, Random Forest helps to de-correlate (decrease overfitting) by making a more stable model and fewer quality issues on predictions and is able to generalize better to data we haven't seen before. By moving from the simpler decision tree to ensembles (Random Forest) as our engagement and acceptance in the app grew and was shown to be more complex with more patterns and interactions in the dataset as it grew, we were able to more accurately and reliably incorporate these potential complex patterns and interactions for accurate and reliable price prediction.



**Fig. 2.1 Price Prediction Flow diagram**



**Fig. 2.2 Flow diagram**

The system's journey begins with a straightforward and secure registration and login process, giving users secure access into the platform. After logging into the platform, users can easily search for nearby certified e-waste facilities. Once a user has inputted all of their details into the system, the Scraply Price Prediction module will use its built-in machine-learning based data analysis, to suggest a fair and reasonable price for each of the user's electronic waste products. The Scraply system is designed to be completely transparent regarding pricing, which will help build trust between the user and the company, while also providing them with additional options when deciding whether to sell an item of electronic waste, or recycle the product at one of the many listed locations. If the user decides to complete a sale, the user will receive, in real-time, updates on the status of their shipment and tracking information on the order. Finally, the user's data will remain private when logout is confirmed.

#### IV. RESULT AND EVALUATION

During the development of the E-Waste Facility Locator, all required functions were performed and a viable method was developed for promoting responsible electronic waste disposal.

**User Registration and Login:** Users can be registered and logged into the system by entering an email address or password; the email verification works correctly and users are able to view their credit points in the system and their history of locations that they have most recently accessed. As shown in Fig. 3.1.

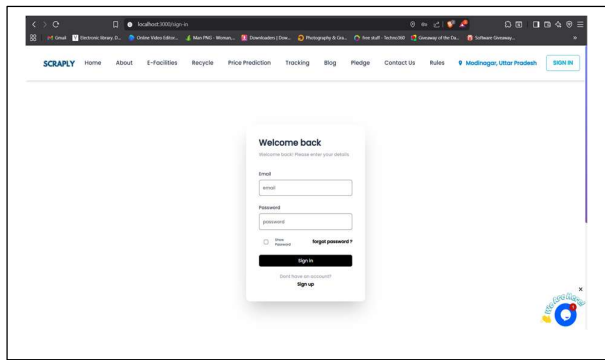


Fig. 3.1 Page for signing up for users

**Facility Search via Map Integration:** The geolocation feature of the platform allowed for accurate identification of users' current locations, displaying a list of certified e-waste collection and recycling facilities in close proximity to their locations. This allowed users to efficiently identify the correct facility without manually searching or too much traveling.

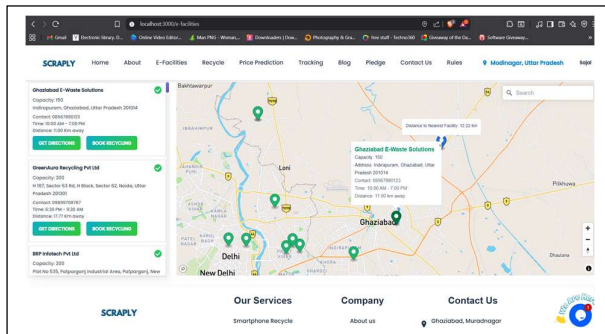


Fig. 3.2 E-waste facility locator

**Educational Pop-Ups and Awareness:** Informational pop-ups provided valuable messaging during strategic actions users completed on the system, such as when they searched for an e-waste facility, or entered details for a device with e-waste. These pop-ups contained valuable messaging about some of the harmful components of e-waste, such as mercury and lead, and how they posed environmental and health risks to the user. Some users indicated that the pop-up insights were new and valuable.

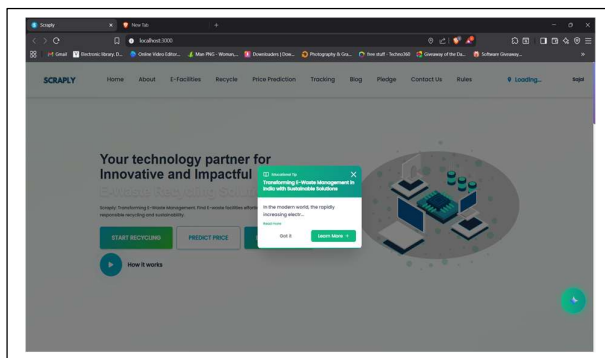


Fig. 3.3 Educational pop ups

**Device Model Input and Reward Points:** Users could input details regarding the old devices they possessed, such

as brand and model, from which the geolocation (approximately) could estimate the amount of recoverable precious metals from the individual e-waste items to automatically award credit points.

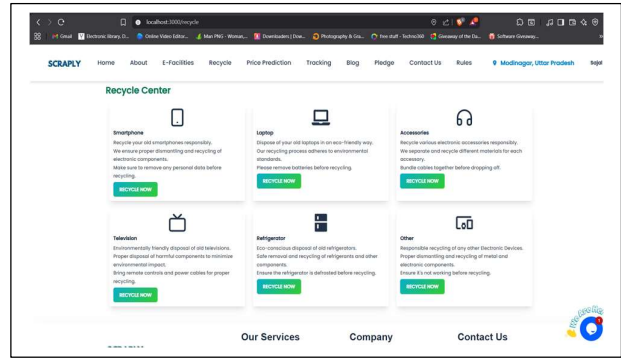


Fig. 3.4 Device categories

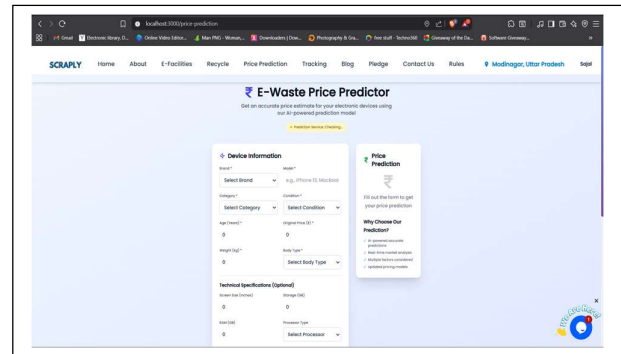


Fig. 3.5 Price Predictor

**Navigation and Directions:** The platform was able to incorporate Google Maps or OpenStreetMap to provide real-time directions from the user's current whereabouts to the selected facility. Step by step navigation of the route locations helped the user plan for the trip.

**Notifications and Events:** The push notification system seems to work well to send the user timely communication about nearby e-waste collection drives, awareness programs, and tips about ways to make use of the system. This keeps users engaged and up-to-date with new information and timelines.

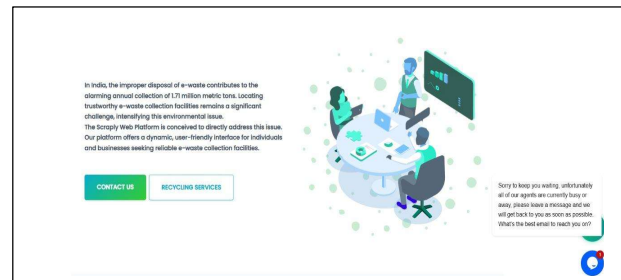


Fig. 3.6 Notifications

**User Feedback and Initial Testing:** The users reported in our first testing report liked the way the system was designed, and the ease of use of the system. Most users said that they found value and were informed. The reward system and educational features were deemed useful too, as it was not

simply about searching for basic features related to a facility. Users liked the clean interface and the map-based experience.

### A. Challenges and Improvements

**Challenges faced:** There were a number of obstacles /challenges encountered during the course of the project. Finding verified and current data of e-waste facilities was difficult, and determining that the location persisted accurately in rural/low-network areas was a challenge. We also needed to create an uncomplicated and engaging interface so that all participants could use it smoothly, as the platform was meant for the general population. Protecting the privacy of the participants was yet another large responsibility. Forecasting prices was complicated and confusing because there were unlimited factors affecting it within dynamic marketplaces. Another obstacle was guaranteeing that the app functioned well on both basic and advanced devices. Making individuals aware of the importance of proper e-waste disposal system was yet another challenge/obstacle to overcome. Finally, engaging with a mixed population of stakeholders and agencies slowed the pace at certain times. Here are some pictures that outline a little of the process during the course of our project in completion, and we hope they make you smile!

**Possible Improvements:** There are many avenues to enhance the project. Additional languages would allow a much larger user base, since many areas are mixed in language. The plan is to have digital wallets or coupons to further assist users redeeming points for rewards. Development of a mobile app would improve and increase access to the platform and make it more user-friendly. Partnering with recycling companies and NGOs would also provide a deeper real-life impact. Finally, there should be the option to resell or donate devices before disposal to preferably reuse and reduce waste further.

## V. CONCLUSION AND FUTURE SCOPE

The "E-Waste Facility Locator" is a technological tool that can reduce the adverse impacts on the environment caused by electronic waste, through providing consumers with easier access to certified recyclers and encouraging reuse (through price prediction) - The E-Waste facility locator facilitates achieving targets outlined in SDG 11 target 11.6 and SDG 12 target 12.5. This application utilizes a circular economy strategy to decrease the volume of e-waste going into landfills and thereby reduces potential health impacts from unsafe disposal of hazardous materials. The locator utilizes technologies such as HTML, CSS, JavaScript, Node.js and MongoDB to provide interactive web-pages and assist in making it easier to find and locate the nearest collection or recycling centers, obtain information about the potential environmental consequences of improperly disposing of electronics and receive points for sharing information about used electronic equipment.

This program was created to be user-friendly and provide information in the best way possible. Proper messaging has been provided regarding the risks associated with improperly disposed of electronics along with the directions to properly recycle them using maps. A points-based plan has been implemented in order to encourage electronic recycling. Users can view their points after logging into their account,

redeeming those points and upon receiving notifications from the website, users will learn about the importance of electronic recycling through pop-up messages. In addition the website will allow users to know when and where local e-waste collection events occur. E-Waste Facility Locator can be so much more than a portal to access your computer and go online. The site could have real time information about local collection centers (i.e., if a center is open or busy). Artificial intelligence can help the site recommend the most efficient route to the recycling center based on distance, wait times, and/or user preference. We can also partner with local municipalities and recycling facilities to expand and improve our data base and provide incentives for people to recycle more via points, coupons, etc. By partnering with the government and developing a payment processing system, we can create a seamless end-to-end recycling process. Overall there is great potential for growth of the locator into a one-stop solution for management of e-waste and truly making an impact on protecting our environment.

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